This is analogous expression for torque experienced by electric dipole in the uniform electric field

$\vec{\tau}_{net} = \vec{p} \times \vec{E}$ which is given in the Unit 1. (Section 1.4.3)

Cases:

(a) When θ = 90°, then the torque on the current loop is maximum which is

$$\vec{\tau}_{net} = abIB\,\hat{j}$$

Note here $\vec{p}_m$ points perpendicular to the magnetic field $\vec{B}$. The torque is maximum in this orientation.

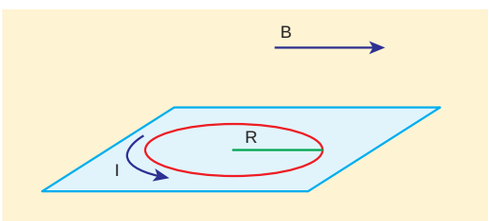
(b) When θ = 0° or 180° then the torque on the current loop is

$$\vec{\tau}_{net} = 0$$

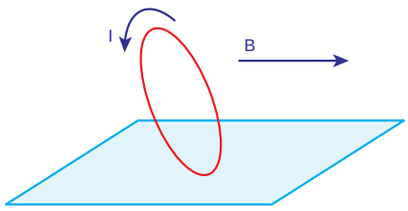
when θ = 0°, $\vec{p}_m$ is parallel to $\vec{B}$ and for θ = 180°, $\vec{p}_m$, is anti - parallel to $\vec{B}$. The torque is zero in these orientations.

EXAMPLE 3.28

Consider a circular wire loop of radius R, mass m kept at rest on a rough surface. Let I be the current flowing through the loop and $\vec{B}$ be the magnetic field acting along horizontal as shown in Figure. Estimate the current I that should be applied so that one edge of the loop is lifted off the surface?



Solution



When the current is passed through the loop, the torque is produced. If the torque acting on the loop is increased then the loop will start to rotate. The loop will start to lift if and only if the magnitude of magnetic torque due to current applied equals to the gravitational torque as shown in Figure

$$\tau_{magnetic} = \tau_{gravitational}$$

$$IAB = mgR$$

But $p_m = IA = I\left(\pi R^2\right)$

$$\pi I R^2 B = mgR$$

$$\Rightarrow I = \frac{mg}{\pi RB}$$

The current estimated using this equation should be applied so that one edge of loop is lifted of the surface.

3.11.2 Moving coil galvanometer

Moving coil galvanometer is a device which is used to indicate the flow of current in an electrical circuit.

Principle When a current carrying loop is placed in a uniform magnetic field it experiences a torque.

Construction

A moving coil galvanometer consists of a rectangular coil PQRS of insulated thin copper wire. The coil contains a large number of turns wound over a light metallic frame. A cylindrical soft-iron core is placed symmetrically inside the coil as shown in Figure 3.66. The rectangular coil is suspended freely between two pole pieces of a horse-shoe magnet.

The upper end of the rectangular coil is attached to one end of fine strip of

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